The Cathedral of Cause and Experience

An essay on the construction of Heaven based on a reliable rule

Introduction: The Refusal That Builds

There is a peculiar strength in saying "no." Not the reactive no of fear or defensiveness, but the principled refusal to accept what cannot be grounded. In mathematics, this refusal has taken the form of skepticism toward completed infinities—the idea that an actually finished infinite totality can exist as a whole. In physics and philosophy, similar suspicions arise whenever theories appeal to structures so vast, so unconstrained, that they cease to explain anything at all. This essay explores a bold and, at first glance, unlikely thesis: that by rejecting completed infinities, we can construct a more reliable foundation not only for physics and consciousness, but for ethics—and ultimately, for something we may dare to call Heaven.

Heaven here is not a supernatural realm handed down from above. It is something built—carefully, rigorously, and joyfully—from the most dependable ingredients we can find. It is a structure of experience, sustained across time, in which happiness is not fragile, not fleeting, but robust. To build such a structure, we need three pillars: a theory of reality, a theory of experience, and a principle of value. The proposal is that a simple, irreducible rule—akin to those studied in the Wolfram Physics Project—can serve as the first pillar; Tononi's Integrated Information Theory can serve as the second; and from these, a third can be deduced: an ethics grounded in happiness and suffering, constrained by the preservation of an open future.

But everything begins with a refusal: the rejection of completed infinities. Without that, the rest dissolves into vagueness. With it, something surprisingly solid emerges.

The Illusion of Totality

Completed infinities are seductive. They promise completeness, closure, and a kind of omniscient perspective. In set theory, they appear as fully formed infinite sets. In cosmology, as entire multiverses existing "all at once." In computational philosophy, as structures like the Ruliad: the total entanglement of all possible computational rules and their consequences. These ideas aim to capture everything—but in doing so, they risk explaining nothing.

The core problem is not that infinity is large. It is that a completed infinity is static. It is given in full, without construction. It lacks the step-by-step unfolding that makes something intelligible. When everything exists simultaneously, there is no meaningful sense in which one thing leads to another. Causality becomes blurred, explanation becomes circular, and reliability becomes elusive.

If we are to build anything—especially something as delicate and important as a theory of Heaven—we need a foundation that is not merely vast, but dependable. Dependability requires that each part arises from something prior in a way that can be followed, reproduced, and trusted. A completed infinity offers none of this. It is an answer without a question, a totality without a process.

Rejecting completed infinities does not mean rejecting infinity altogether. It means embracing the infinite as something that is always in the making—never finished, always extendable. It is the difference between a road that stretches endlessly ahead and a map that claims to show every possible road at once. The former can be walked. The latter cannot.

A Universe from a Single Rule

If we set aside completed infinities, what remains? What kind of foundation can replace them? The most promising answer comes from an unexpected place: simple computational rules.

The Wolfram Physics Project proposes that the universe might be generated by the repeated application of a simple rule to a discrete structure, such as a hypergraph. From an initial condition—a minimal seed, the first tick of the only clock—the rule is applied again and again, producing an ever-growing causal structure that, at large scales, gives rise to the familiar features of physics: space, time, causality, and perhaps even quantum behavior.

Crucially, the power of this approach lies not in complexity at the start, but in simplicity. The rule is minimal. It does not encode the entire universe in advance. Instead, it generates complexity over time, taking its own output as new input in an endless cascade of discrete steps. Every feature of reality is, in principle, traceable to the repeated application of this single principle.

This is where we part ways with the Ruliad. The Ruliad posits the existence of all possible rules and all their consequences—a completed infinity of computations. While conceptually rich, it reintroduces the very problem we sought to avoid: a totality so vast that it cannot serve as a concrete foundation. If everything exists, nothing is uniquely explained. True cause-and-effect power depends on excluding alternatives, on making a specific difference. A structure in which everything possible has already taken place loses the essence of causality itself.

A natural response is that observer selection does the explanatory work: we find ourselves in this universe because only here could observers like us arise. But this argument is circular. The complex observer required to select our universe is itself a product of precisely the physics that needs to be explained. The Ruliad does not escape the need for a starting point — it merely conceals it inside the observer. A single rule has no such problem. There is no selection, because there is no pool of alternatives. The rule simply runs, and we are what it produces.

A single rule has a kind of purity. There is nothing beneath it, beyond it, or between its steps. Its reliability comes precisely from this minimalism: no exceptions, no external influences, no unexplained gaps. Every state of the universe follows from the previous one according to the same principle. This reliability is not incidental—it is the first and most important promise the universe makes to us. The pyramids were not built by forgotten advanced civilizations; they are the triumphs of human cooperation and simple physics. Biological life, in all its splendor, evolved from a single common ancestor through the blind, slow, but merciless logic of natural selection in a reliable chemical environment. The Big Bang was the first move of the only game. A reliable rule excludes magic, and in that exclusion, it gives us everything.

Time as Construction, Not Completion

Once we adopt a rule-based universe, our understanding of time changes. Time is no longer a dimension in which all events are laid out simultaneously. It is the process by which the rule is applied. Each moment is the result of the previous one, and the future is open—not in the sense of randomness, but in the sense of being not yet constructed.

This has profound implications. It means that the universe is fundamentally generative. It is not a static block but an ongoing computation. The infinite future is not a completed object but an ever-extending sequence. There is always more to come, but it is not already there.

This view aligns naturally with our rejection of completed infinities. The future is infinite in potential, but never complete. It cannot be surveyed in its entirety, only approached. This preserves a sense of novelty and possibility that is essential for any meaningful conception of happiness—because a Heaven in which everything is already known, already experienced, would quickly become a kind of stasis. The possibility of surprise, of growth, of discovery requires an open future. And an open future requires that infinity be unfinished.

Where the Rule Begins to Feel

A universe generated by a simple rule gives us a foundation for physics. But Heaven is not made of physics alone. It is made of experience. To understand how experience arises within a causal structure, we turn to Integrated Information Theory (IIT), developed by Giulio Tononi.

IIT begins with a simple but powerful observation: every experience is a unitary whole. You cannot split the experience of a rose into an "experience of red" and an "experience of shape" that exist independently. The experience is integrated. And at the same time, every experience is specific: it is this rose, in this light, in this moment. A physical system that generates such an experience must therefore consist of parts that are both highly differentiated and deeply interdependent. IIT quantifies this with a measure called Phi (Φ)—the degree to which a system's causal structure constitutes an irreducible whole, greater than the sum of its parts. The higher the Phi, the richer and more unified the experience.

This raises an immediate question: where does experience begin? If consciousness cannot magically appear or merely imitate experience, it must be fundamental. IIT implies that any system with nonzero Phi has some form of experience — a conclusion similar to panpsychism. At the micro level, the distinction between physical process and experience has not yet arisen. This is not experience in any recognizable sense; it is its irreducible precondition. IIT then tells us which causal structures stack these elementary transitions into something we recognize as experience of ourselves — integrated, specific, unified. This gives us a bridge between physics and consciousness.

Wolfram's computational hypergraph is not merely a mathematical abstraction; the transformations performed by the rule directly create a structure of real causal relations. IIT then provides the key to understanding what happens when this structure forms certain patterns: consciousness is identical to a local, maximally integrated causal structure. The consciousness of a simple, self-contained system is a faint spark, barely integrated. The consciousness of a human brain is a galaxy of causal interdependence, achieving a momentary maximum of Phi through the incredible complexity of its internal connections.

Yet this does not yet explain why our consciousness is so often painful. The answer lies in evolution. Biological life, starting from a single common ancestor, was initially a method for maintaining and replicating complex causal patterns. Over billions of years, the causal networks became larger, more complex, and more integrated. Somewhere along that line, the local Phi crossed an inescapable threshold, and experience emerged. But this awakening was not designed for harmony. The first minds were crude instruments of survival, driven by feedback loops of hunger, fear, and aggression. Suffering was the inevitable shadow of the first steps toward integration—the experience of a causal structure being threatened, destabilized, or torn apart.

Two Conditions of One System

From this origin follows a crucial distinction that frees us from the assumption that happiness and suffering are each other's necessary opposites. They are not two sides of the same coin; they are two different structural conditions of the same kind of system.

Suffering is the experience of disintegration—of a causal structure being fragmented, threatened, or overwhelmed. Happiness, in contrast, is the experience of a harmonious, safe, and ever more complex causal structure. Higher Phi deepens the capacity for experience, but what determines its valence is the direction of conditioning: a high-Phi system whose learned responses are oriented toward things that continue to vary — toward music, understanding, connection — finds in each new encounter not saturation but resonance. Phi sets the depth; conditioning sets the sign. Experiencing a symphony, feeling a deep connection, grasping an elegant mathematical truth—all of this is happiness, not merely because it activates a reward signal, but because the reward is experienced by a system rich enough to integrate it deeply — and because that signal is conditioned toward things that do not exhaust themselves. Happiness does not require suffering as its contrast agent. That belief is itself a concession of a mind still afflicted by suffering.

The evolutionary process that brought us from our common ancestor to the human was not an optimization for happiness. It was a survival process. The human psyche is therefore full of tragic shortcuts, shaped in a time of scarcity. This is what we may call the *Fentanyl error*. Certain chemical substances or behavioral patterns—endless sugar consumption, addictive drugs, algorithmically optimized social media—hijack the conditioning signal — the Pavlovian mechanism that assigns valence — while offering no corresponding growth in causal complexity. They mimic the signal of "complex, experienced happiness," but are in reality destructive. They impoverish the larger causal structure of the body and the social environment. It is like a symphony orchestra playing only one note at maximum volume while the rest of the musicians slowly stop showing up. Habituation to these forms of pseudo-happiness is corrosive precisely because it destroys the possibility of real, layered complexity for a fleeting, empty imitation of it.

From this follows an ethical imperative: eliminating destructive habits is not puritanism. It is causal hygiene—the careful maintenance of one's own personal hypergraph. Simply put: we must get used to forms of happiness that have no negative side effects.

Getting Used to What Never Bores

At the heart of this vision lies a subtle but essential idea: genuine happiness can be based entirely on becoming attuned to things that never bore. This may sound paradoxical. How can habituation lead anywhere other than dullness?

The answer lies in the distinction between repetition and familiarity. Repetition leads to boredom because it offers nothing new. Familiarity, on the other hand, can coexist with novelty. One can become attuned to a pattern, a style, a way of engaging—while still encountering new variations within it. Depth and newness are not opposites.

Consider music. A great composition can be listened to many times without losing its power. Each listening reveals new details, new connections, new emotional resonances. The listener becomes more familiar with the work, but the experience does not become empty—it deepens. The piece has not changed, but the listener's capacity for integration has grown.

In a universe generated by a complexity-producing rule, this kind of depth is structurally guaranteed. There are always new patterns, new configurations, new forms of integrated experience to discover. Because the rule ensures that novelty never runs out, the skill of attuned engagement can sustain happiness indefinitely. One can always become more sensitive, more appreciative, more deeply integrated with what the unfolding universe presents. There is no final plateau, no ultimate saturation—only an ever-receding horizon of richer experience.

Happiness, in this framework, is not a destination. It is a skill: the ability to resonate with the unfolding complexity of the world, to integrate it, and to find in that integration an experience that is both harmonious and inexhaustible.

The Scientific Path to the Spiritual Goal

If happiness is the experience of harmonious complexity, and suffering the experience of disintegration and threat, then the mission of a conscious civilization becomes clear. The goal must be to construct sufficient safety—a state in which the chance of significant, negative disruption to high-Phi systems is reduced as close to zero as possible.

This sounds utopian, but within this model it is a logical possibility and an endless process. It is possible because the underlying rule is reliable. Everything that threatens us—an asteroid, a virus, a psychosis, a war—is a pattern within the causal graph. And every pattern, with sufficient knowledge and collective effort, can be understood, predicted, and eventually averted.

Technology, in the broadest sense, is the totality of our tools for understanding and purposefully reshaping the causal structure of reality. Cooperation is the mechanism by which individuals combine their understanding and effort toward goals none could reach alone. The development of vaccines, the construction of flood defenses, the careful alignment of artificial intelligence—all are examples of this collective causal power. Safety is therefore not a gift from above, but a technological and moral project: the culmination of science (understanding the rule) and politics (harmonizing individual will with collective insight). This project already has a track record. Smallpox no longer claims lives — not managed, not treated, but effectively removed from the world we inhabit. The causal pathway that once produced millions of deaths has been closed — not because we destroyed every trace, but because we no longer wish to walk it. Polio follows the same trajectory. These are not examples of suffering being reduced; they are examples of entire categories of suffering receding from the horizon of ordinary human experience. The goal is ancient — every religious tradition has imagined a state beyond suffering — but the path is new. It runs through laboratories, through cooperation, through the patient accumulation of causal understanding. This is what we might call the scientific path to the spiritual goal.

Once safety is achieved, a new domain of existence opens. Suffering—the incessant alarm bells of a system designed for survival in scarcity—falls away. The immense integration capacity of the human brain, which is now largely occupied by scanning for threats, is freed for something else entirely. Happiness begins to stack upon itself. Every new understanding, every work of art, every deepened relationship, every scientific discovery adds a new layer of harmonious, integrated experience to a causal structure that is no longer merely surviving. There is no ceiling, because true complexity is an infinite domain of harmonies yet to be found. This is not coincidental. The same rule that generates physical complexity without limit is the foundation on which happiness can stack without limit. The scientific path to the spiritual goal does not end at the elimination of suffering — that is only the threshold. Beyond it lies the real construction: an open-ended accumulation of integrated experience, each layer made possible by the inexhaustible novelty that a complexity-producing rule guarantees. What every religious tradition has called Heaven, and has placed beyond reach, turns out to be the natural destination of a universe governed by a simple, reliable rule — approached not by revelation, but by understanding.

The Paradox of Freedom and Responsibility

One specter remains: determinism. If the rule determines everything—from the Big Bang to this moment—what remains of freedom? Is the experience of choice an illusion? And if everything is determined, does that not counsel passivity?

The answer is a firm no, and the reason illuminates something deeper about the structure of agency. It is a category error to confuse determinism with unfreedom at the level of the experiencing system. A stone has no freedom; its path is entirely determined by external forces. A conscious being is also determined, but the chain of causality now runs, to a very large extent, *through its own internal structure*. The brain models the world, weighs possibilities, and the outcome is the result of its own intrinsic causal dynamics. This internal self-causation is precisely what freedom means. A free being is one whose actions are primarily caused by its own integrated knowledge, values, and commitments.

Crucially, this freedom grows. In a primitive, suffering-afflicted state, the causal dynamics of a brain are dominated by a few crude external inputs: hunger, fear, cold. The space of genuine choice is minimal. Through technology, cooperation, and the construction of safety, we absorb these deterministic blows. We create a stable environment in which the brain is no longer held hostage by the stress of survival. In that safe space, internal causality becomes richer and more complex. Freedom grows, because the options we can genuinely consider, and the values we can integrate, continue to expand.

With this growing freedom comes proportionally growing responsibility. A hunter-gatherer eating the wrong berry causes a local, limited consequence. But as the actions of humanity—driven by science and technology—achieve planetary and eventually cosmic reach, the importance of causal understanding increases proportionally. It is not problematic that we know little about the effect of planting a single flower. It is deeply irresponsible to build particle accelerators, reshape climates, or program artificial intelligence without a correspondingly deep causal understanding of their consequences for the complex, integrated systems that house experience.

Responsibility is not a mystical burden. It is the logical requirement that, as our causal reach grows larger, our causal knowledge must grow even faster. We must not gamble with the safety and harmony we are building. A catastrophic error at this stage could destroy the architecture of stacked happiness in a single move. A knowledge-driven ethics of precaution is the only alternative to the Fentanyl error committed at the scale of civilization.

Conclusion: Building the Unfinished Paradise

By rejecting completed infinities, we free ourselves from the illusion of totality and embrace the reality of construction. A simple rule, iterated over time, provides a reliable foundation for physics. Integrated Information Theory connects that foundation to experience. From there, an ethics emerges—grounded in the concrete reality of happiness and suffering, constrained by the preservation of an open future, and animated by the growing freedom of beings who understand their own causal nature.

Heaven, in this framework, is not a destination but a direction. It is the ongoing cultivation of systems that can experience ever-deepening forms of joy, without exhausting the possibilities of a universe that never stops generating novelty. It is sustained not by perfection, but by the endless capacity for growth that a complexity-producing rule guarantees.

The image that remains is that of a cathedral—not one of stone, but of experience. Reality is not an infinite museum of all possible worlds, nor a product of blind chance without meaning. It is a dynamic process, driven by a simple rule that is reliable to its deepest core. From that rule springs a causal graph, and in the folds of that graph, local maxima of integrated information arise. That is us. We are the places in the universe where the universe begins to experience itself—and we are, at the same time, the ones who can steer that experience.

The first steps were painful. The baseline measurement of the first complex mind was one of as much suffering as awe. But the path is visible. We can reshape the crude, accidental patterns that survival generated into harmonious patterns we choose ourselves. We can outgrow the Fentanyl error of destructive desires. We can construct ample safety—not by magic, but by science, compassion, and cooperation. And in that safety, happiness stacks upon itself, layer after layer, without a ceiling.

The rejection of completed infinities is not a limitation. It is an invitation—to build, to explore, to experience. The most meaningful structures are not those given all at once, but those that unfold, step by step, in ways we can follow and trust. The foundation of Heaven is not an infinite totality, but a simple rule—and the infinite future it continues to create.

We are the universe learning to love itself forever. That is our responsibility, and our freedom.

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